If $\lambda_1 \neq \lambda_2$, let $z = xy^{-1}$. Then the last term of the summation is

$$(q^2+q)\sum_{x\neq y}\lambda_1(z)\overline{\lambda_2(z)} = (q^2+q)\sum_{x;z\neq 1}\frac{\lambda_1}{\lambda_2}(z) = (q^2+q)(q-1)\sum_{z\neq 1}\frac{\lambda_1}{\lambda_2}(z).$$

Since

$$\sum_{z\in\mathbb{F}_q^\times}\frac{\lambda_1}{\lambda_2}(z) = 0,$$

because the sum of all roots of unity of a given order $m > 1$ is zero, the last term becomes

$$-(q^2+q)(q-1)\frac{\lambda_1}{\lambda_2}(1) = -(q^2+q)(q-1).$$

The difference between this case and the case of $\lambda_1 = \lambda_2$ is equal to

$$(q^2+q)[(q-2)(q-1)+(q-1)] = |G|,$$

so this is an irreducible representation by Lemma 5.7.2.

To prove the third assertion of the theorem, we look at the characters on hyperbolic elements and note that the function

$$\lambda_1(x)\lambda_2(y) + \lambda_1(y)\lambda_2(x)$$

determines $\lambda_1, \lambda_2$ up to permutation. □

The representations $W_\mu, V_{\lambda_1,\lambda_2}$, $\lambda_1 \neq \lambda_2$ are called principal series representations.

**5.25.4. Complementary series representations.** Let $\mathbb{F}_{q^2} \supset \mathbb{F}_q$ be a quadratic extension $\mathbb{F}_q(\sqrt{\varepsilon}), \varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_q^2$. We regard this as a 2-dimensional vector space over $\mathbb{F}_q$; then $G$ is the group of linear transformations of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$. Let $K \subset G$ be the cyclic group of multiplications by elements of $\mathbb{F}_{q^2}^\times$,

$$K = \{\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix}\}, \qquad |K| = q^2 - 1.$$

For $\nu : K \to \mathbb{C}^\times$ a homomorphism, let

$$Y_\nu = \mathrm{Ind}_K^G\, \mathbb{C}_\nu.$$